Assignment # 02

developing a simple AI system to analyze student performance and predict results

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4th semester

AB_305 Programming For AI
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REG NO: 24PWAI0038
CLASS SECTION: A

"On my honor, as a student of University of Engineering and Technology,I have neither given nor received unauthorized assistance on this academic work."
Student Signature ______________________

Submitted to:

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Date: 10 Mar 2026

NumPy for AI - Mini Project Assignment

Objective:

This project will help students understand how NumPy is used in Artificial Intelligence for data handling, preprocessing, and basic prediction.

Project Scenario:

You are developing a simple AI system to analyze student performance and predict results.

Dataset Format:

Maths	Science	English	Computer	Result
87	91	94	83	1
43	41	84	23	0
56	76	78	59	1
76	89	91	99	1
27	19	34	45	0
48	19	49	38	0
83	78	67	54	1
28	23	14	12	0
62	77	57	67	1
70	80	90	100	1

Tasks:

- Create a dataset of at least 10 students using NumPy.
- Separate features (X) and labels (y).
- Calculate average, maximum, and minimum marks.
- Find number of students who passed and failed.
- Identify students scoring above 80 and below 50.
- Normalize the dataset.
- Apply simple prediction logic.
- Calculate accuracy of predictions.

Python Code:

```
import numpy as np
#creating dataset of 10 student
dataset = np.array([
  [87,91,94,83,1],
  [43,41,84,23,0],
  [56,76,78,59,1],
  [76,89,91,99,1],
  [27,19,34,45,0],
  [48,19,49,38,0],
  [83,78,67,54,1],
  [28,23,14,12,0],
  [62,77,57,67,1],
  [70,80,90,100,1]
])

# separating input and output features
x = np.array(dataset[:,0:4])
y = np.array(dataset[:,4])

# average, maximum, and minimum marks
average_marks = np.mean(x)
print("Average Marks :",average_marks)
minimum_marks =np.min(x)
print("Minimum Marks :",minimum_marks)
maximum_marks = np.max(x)
print("Maximum_marks :",maximum_marks)

#number of students who passed and failed
pass_student = np.sum(y==1)
print("Pass Students :",pass_student)
```

```
fail_student = np.sum(y==0)
print("Fail students :",fail_student)

#Identify students scoring above 80 and below 50
print("Student scored above 80 :",x[x>80])
print("Student scored below 50 :",x[x<50])

#Normalize the dataset
normalizing = (x-np.min(x) )/ (np.max(x)-np.min(x))
print("Normalized data :",normalizing)

# Apply simple prediction logic
average_per_student = np.mean(x,axis=1)
prediction = ( average_per_student >= 60).astype(int)
print("Average marks of per student :",average_per_student)
print("Prediction :",prediction)
print("Actual :",y)

# accuracy

accuracy = np.mean(prediction==y)
print("accuracy :",accuracy)
```

Output Of The code:

```
... Average Marks : 60.275
    Minimum Marks : 12
    Maximum_marks : 100
    Pass Students : 6
    Fail students : 4
    Student scored above 80 : [ 87  91  94  83  84  89  91  99  83  90 100]
    Student scored below 50 : [43 41 23 27 19 34 45 48 19 49 38 28 23 14 12]
    Normalized data : [[0.85227273 0.89772727 0.93181818 0.80681818]
     [0.35227273 0.32954545 0.81818182 0.125     ]
     [0.5        0.72727273 0.75       0.53409091]
     [0.72727273 0.875      0.89772727 0.98863636]
     [0.17045455 0.07954545 0.25       0.375     ]
     [0.40909091 0.07954545 0.42045455 0.29545455]
     [0.80681818 0.75       0.625      0.47727273]
     [0.18181818 0.125      0.02272727 0.        ]
     [0.56818182 0.73863636 0.51136364 0.625     ]
     [0.65909091 0.77272727 0.88636364 1.        ]]
    Average marks of per student : [88.75 47.75 67.25 88.75 31.25 38.5  70.5  19.25 65.75 85.  ]
    Prediction : [1 0 1 1 0 0 1 0 1 1]
    Actual : [1 0 1 1 0 0 1 0 1 1]
    accuracy : 1.0
```

Explaination of learning:

1. After That assignment I learned that how numpy is used in python and how it work with arrays by creating the array of 10 students.
2. Then I learned how to separate features using numpy.
3. Then I learned the simple statistical operation of numpy like student average marks,maximum marks and minimum marks.
4. Another important concept I learned is data filtring ,that is I find the students scoring above 80 and below 50 using condition in numpy.which demonstrate how AI categorized data automatically.
5. Another important concept that I learned is data normalization ,which bring all the data into similar and small scale due to which the accuracy and prediction of AI model become much better.
6. And finally I applied simple prediction logic to predict the result using average marks and then compared the actual with prediction to calculate accuracy.

This project give me experienced using numpy in simple artificial intelligence task.

================================= The End =================================